Active sensing explains cortical variability and improves visual information

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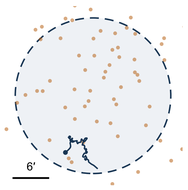
Fixational eye movements: noise or signal?

Even during fixation the eyes drift and make microsaccades, so each repeat of an image lands on a different part of the foveal receptive fields.

Noise view
A repeated stimulus is a repeated input, so trial-to-trial variability is noise to be averaged away.

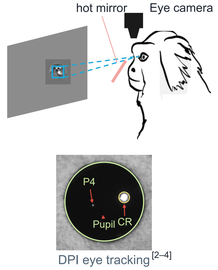
Active-sensing view
Eye motion turns space into time and is part of how the fovea encodes fine detail[5,6].

Q1 How much V1 "noise" do fixational eye movements drive? → 1–5
Q2 Does that activity carry visual information? → 6–8



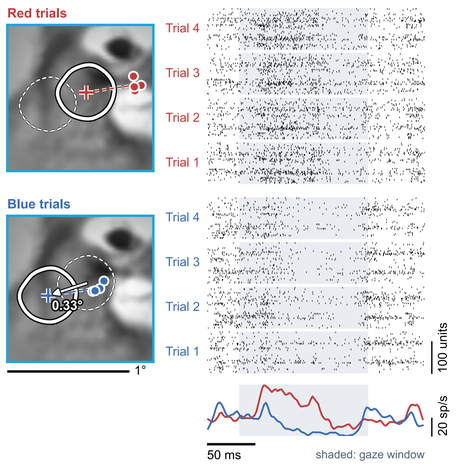
Median gaze on 88 repeats (navy: one path) vs a typical foveal V1 RF (0.42°)

1 Tiny gaze shifts reshape population responses



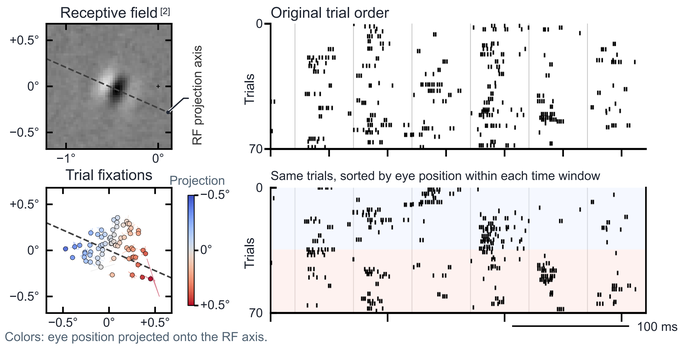
DPI eye tracking[2–4]

Gaze on one trial
Population RF on these trials (same neurons in both views)
Where it sat on the other trials
Views: the cyan screen patch, enlarged



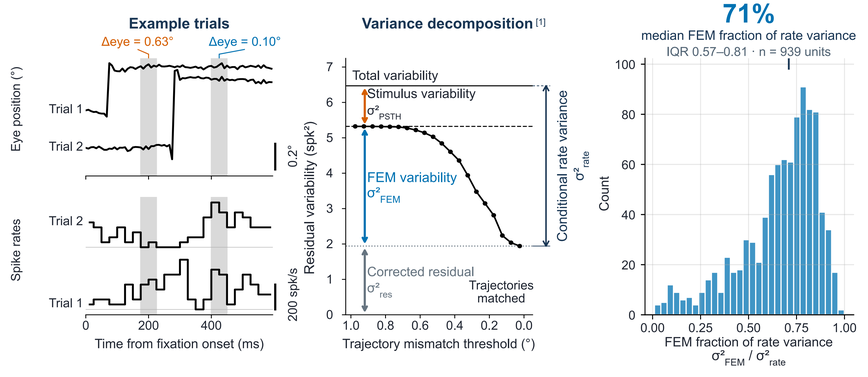
Gaze shift 0.33° — smaller than a typical receptive field (median diameter 0.42°, 96 units)
Dots: each trial's median gaze in the shaded window. RF shape from noise mapping, located from these trials.
One session, 132 units; same image sequence on every trial. Lines: mean rate per unit.
Dataset: 2 marmosets · 1,022 foveal V1 units · 19 sessions · 20 Hz natural images

2 One neuron's "noise" tracks the eye

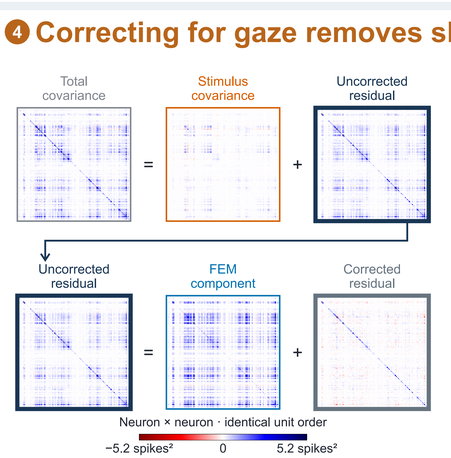


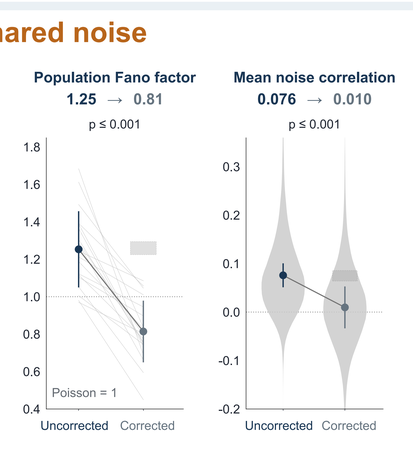
Colors: eye position projected onto the RF axis.

3 Eye movements drive most rate variance (71%)



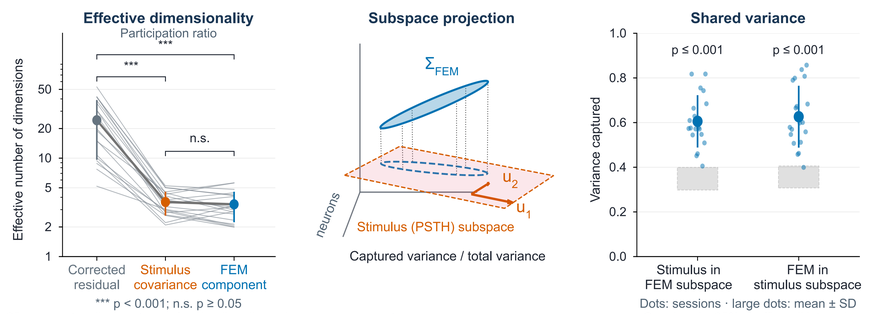
4 Correcting for gaze removes shared noise





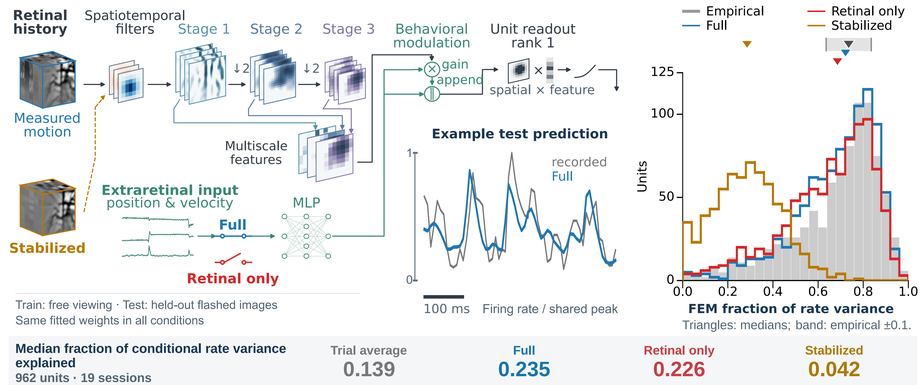
19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

5 Eye-driven variability follows stimulus dimensions



*** p < 0.001; n.s. p ≥ 0.05
25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

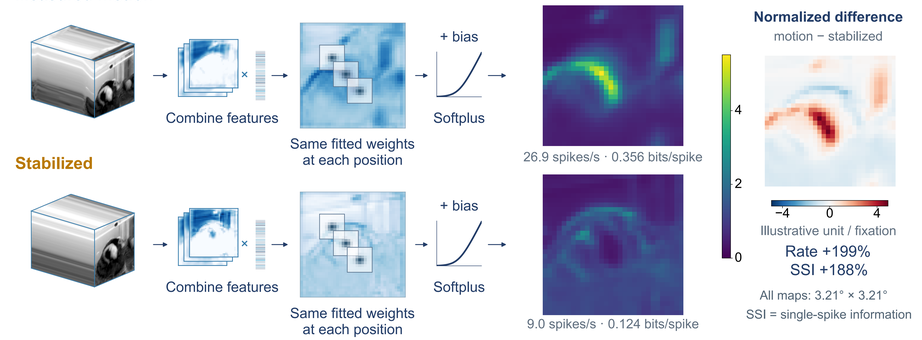
6 The retinal image, not an eye signal, predicts single trials



Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions
Triangles: medians; band: empirical ±0.1.

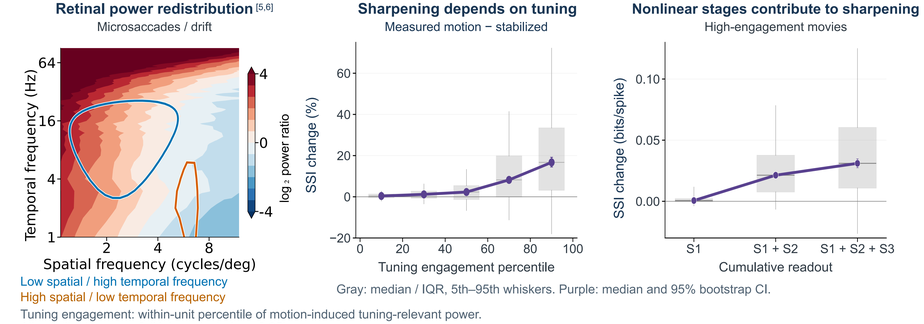
Median fraction of conditional rate variance explained 962 units · 19 sessions	Trial average	Full	Retinal only	Stabilized
	0.139	0.235	0.226	0.042

7 In the model, eye motion sharpens spatial coding



8 Sharpening grows with tuning and model depth

725 units



Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).
Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

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